

Coding & Solution

Date	26 March 2026
Project Name	Smart Lender

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This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements. Instructions: 1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack. 2. The solution should address the core problem statement and implement the key functional requirements. 3. Include all source files, configuration files, and setup instructions needed to run the application. 4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sandeepemmanuel10-blip/Smart-Lender
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Bootstrap
Key Features Implemented	User-friendly loan application form, applicant data validation, machine learning-based loan approval prediction, XGBoost model integration, real-time prediction results, responsive web interface, IBM Cloud deployment support
Pending / Incomplete Features	User authentication, loan application history, admin dashboard, email/SMS notifications, database integration for storing applicant records

Setup / Run Instructions	1. Clone the GitHub repository. 2. Install the required dependencies using pip to install -r requirements.txt. 3. Run the Flask application using the Python app.py. 4. Open the browser and visit. 5. Enter applicant details and click Predict to view the loan approval results.
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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender application successfully implements a machine learning-based loan approval prediction system using Flask and the XGBoost algorithm. The code follows a modular structure with separate components for data preprocessing, model loading, prediction, and the web interface. The application provides accurate, real-time loan approval predictions and can be further enhanced by adding user authentication, database integration, and advanced reporting features.